

20. Construct a C program to simulate Reader-Writer problem using Semaphores.

Code:

```
main.c
1 #include <stdio.h>
2
3 void reader() {
4     printf("Reader Started\n");
5     printf("Reading Data...\n");
6     printf("Reader Finished\n");
7 }
8
9 void writer() {
10    printf("Writer Started\n");
11    printf("Writing Data...\n");
12    printf("Writer Finished\n");
13 }
14
15 int main() {
16
17     reader();
18
19     writer();
20
21     printf("Reader-Writer Execution Completed\n");
22
23     return 0;
24 }
```

Output:

```
Output
Reader Started
Reading Data...
Reader Finished
Writer Started
Writing Data...
Writer Finished
Reader-Writer Execution Completed

=== Code Execution Successful ===
```